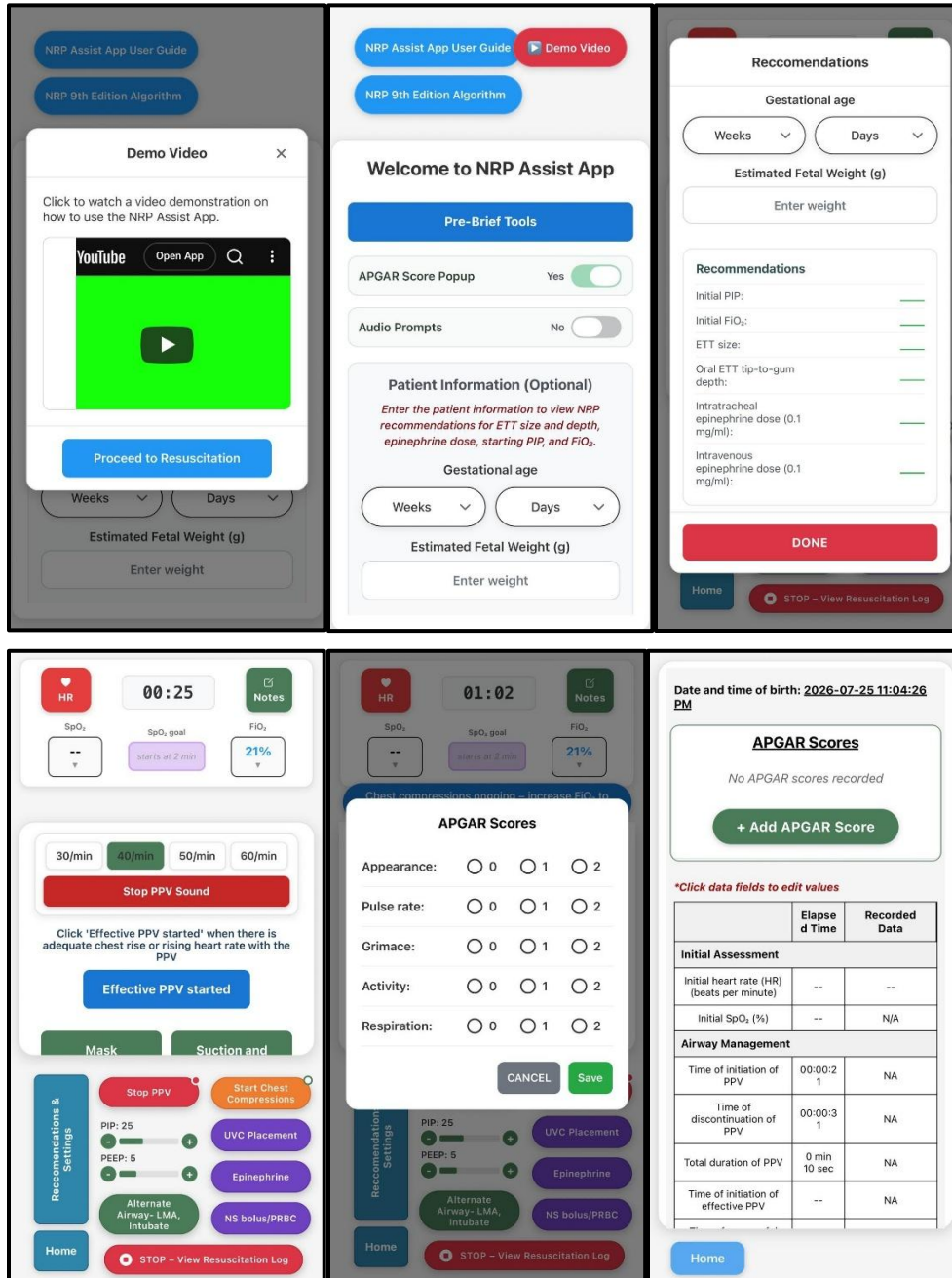


NRP Assist app Investigator Brochure:

Screenshots from the NRP Assist App:



[Link to the App](#)

1. Introduction

The NRP Assist application is an iOS-based decision-support tool designed to assist neonatal resuscitation providers by offering time-based prompts and reference information consistent with the Neonatal Resuscitation Program (NRP) 9th Edition guidelines¹ (2025). The app is intended to support cognitive organization and task tracking during neonatal resuscitation but is not intended to replace clinical judgment or formal NRP training.

The application is intended for use during NRP simulation and is not intended for real-life clinical use. It is essential to emphasize that this app is not a substitute for comprehensive medical training or certification in resuscitation techniques. Users must not rely on this app as their sole source of knowledge.

2. Purpose

The NRP Assist app is designed to:

- Support adherence to the NRP 9th Edition neonatal resuscitation algorithm.
- Reduce cognitive load during neonatal resuscitation simulations.
- Improve team organization and task tracking.
- Facilitate documentation of resuscitation events during simulation scenarios.

3. Key Features

- Timers & Alerts:
 - Automatically initiates timers at key resuscitation milestones.
 - Provides time-based prompts consistent with NRP recommendations.
- Apgar Prompts:
 - Displays reminders at 1, 5, 10, 15, and 20 minutes.
 - Allows real-time entry of individual Apgar score components.
- Oxygen Saturation Reference:
 - Displays expected preductal SpO₂ targets according to postnatal age.
 - Provides prompts to adjust FiO₂ based on recorded oxygen saturation.

- Epinephrine Dose Calculator:
 - Calculates recommended intravenous and endotracheal epinephrine doses based on the estimated fetal weight entered by the user.
- Event Recording:
 - Records interventions throughout the resuscitation process
 - Automatically generates an editable **Resuscitation Log (PDF)** summarizing recorded events.

4. Intended Users

NRP Assist is intended for healthcare professionals who are trained and credentialed in NRP, including physicians, nurses, advanced practice providers, physician assistants, respiratory therapists and other NRP-trained neonatal resuscitation providers.

5. Getting Started

Step 1. Launch the App

Open the NRP Assist app on a compatible iPhone or iPad.

Step 2. Review the Disclaimer

After accepting the disclaimer, a demonstration video is displayed. The video may be viewed immediately or closed and accessed later using the Demo Video icon in the upper-right corner of the Home screen.

The Home screen also provides access to:

- NRP Assist User Guide
- Latest NRP Algorithm
- Pre-Brief Tools

Step 3. Start the Scenario

Tap Birth to begin the simulation.

The Main Screen opens and is divided into three interactive sections:

- Upper Section
- Middle Section

- Lower Section

6. Home Screen

The Home screen provides access to pre-brief resources and scenario settings.

Pre-Brief Tools

The Pre-Brief Tools include:

Four Pre-birth Questions

A reminder of the four standard NRP pre-birth briefing questions.

Team Role Assignment

The application includes editable role assignments for teams consisting of three or four members.

Users may:

- Edit task assignments
- Move tasks between team members
- Add additional team members
- Save changes for the current session or all future sessions
- Restore the default role assignments at any time

Equipment Checklist

The equipment checklist is based on the **NRP 9th Edition** recommendations.

Users may:

- Mark equipment as available
- Uncheck items as needed
- Reset the checklist at any time

Recommendations and Settings:

Before beginning the simulation, users may:

- Enter estimated fetal weight
- Enter estimated gestational age
- Enable or disable:
 - Apgar reminders
 - Audio prompts

Entering patient information generates recommendations for:

- Initial FiO_2
- Initial PIP
- Endotracheal tube size
- Endotracheal tube depth
- Intravenous epinephrine dose
- Endotracheal epinephrine dose

If patient information is not entered initially, the application will prompt the user before an epinephrine dose is calculated.

Users can return to the Home screen from any point during the simulation.

7. Main screen:

The main screen has 3 sections which interact with each other.

Upper Section

The Upper Section allows users to record patient information while providing real-time reference information.

- It displays the time elapsed since start of resuscitation in the center of the section

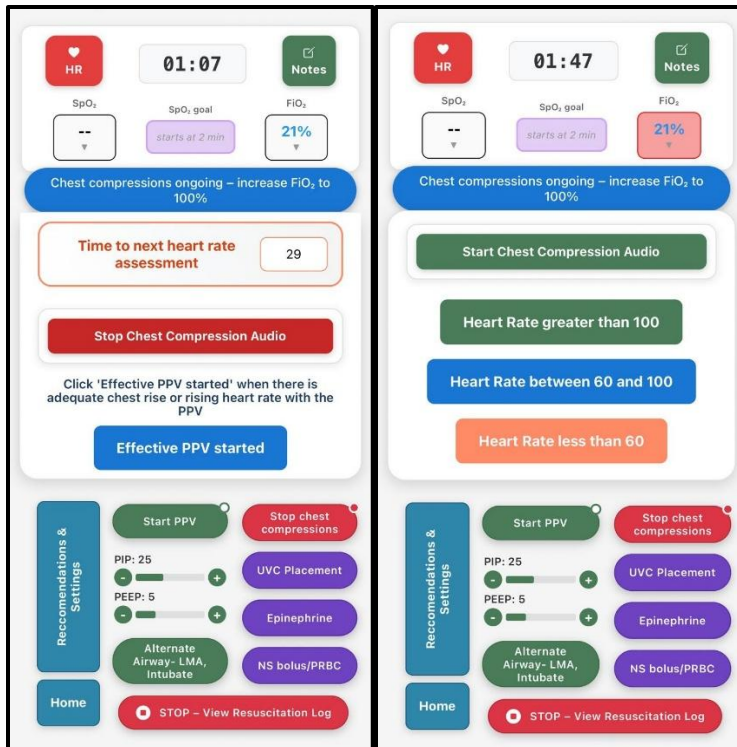
The screenshot shows the NRP Assist App interface. At the top, there are three buttons: 'NRP Assist App User Guide' (blue), 'Demo Video' (red), and 'NRP 9th Edition Algorithm' (blue). Below these is a 'Welcome to NRP Assist App' section. Underneath is a 'Pre-Brief Tools' section with two toggle switches: 'APGAR Score Popup' (set to 'Yes') and 'Audio Prompts' (set to 'No'). Below that is a 'Patient Information (Optional)' section with a note: 'Enter the patient information to view NRP recommendations for ETT size and depth, epinephrine dose, starting PIP, and FiO_2 .' This section contains two dropdown menus for 'Gestational age' (with 'Weeks' and 'Days' options) and a text input field for 'Estimated Fetal Weight (g)' with the placeholder 'Enter weight'.

- It facilitates recording of vitals including heart rate, SpO2 and FiO2.
- The scribe notes on the right upper corner allows users to enter additional narrative information and automatically records timestamps for major interventions, including intubation, chest compressions, epinephrine administration, normal saline boluses, PRBC administration. This information can be viewed anytime by opening the scribe notes.
- The upper section also displays SpO2 goals per age and prompts users to adjust FiO2 based on the SpO2 entered.

Middle Section

The Middle Section provides assessment, decision support, and time-based guidance. The features include:

- Heart rate assessment: Users may select heart rate categories (>100, 60–100, or <60), or enter the heart rate directly in the upper section. When a heart rate is entered, the application automatically selects the appropriate category and updates the guidance.
- Visual guidance: Displays real-time prompts based on recorded vital signs and interventions.
- Secondary timers: Displays countdown timers indicating when reassessment or subsequent interventions should occur.
- Positive Pressure Ventilation (PPV) Guidance: When PPV is clicked in the lower section, the middle section shows the option to start/stop the PPV audio guidance, as well as the option to adjust the audio to 30, 40, 50 or 60 breaths/minute. It also displays the ventilation corrective steps. Selecting Pressure Increase allows the user to record a new PIP, which can also be done directly in the lower section. Selecting Alternate Airway opens options for LMA and intubation. The app records the duration of each airway attempt. The alternate airway can also be operated from the lower section.
- Chest Compressions Guidance: When chest compressions are initiated, controls for chest compression audio guidance become available.
- APGAR Score Prompts: Users may enter all or selected components of the Apgar score or dismiss the prompt and continue the resuscitation scenario.



Lower Section:

The Lower Section contains action buttons used to record interventions performed during the simulation. These include:

- Positive pressure ventilation
- Chest compressions
- Epinephrine administration
- Normal saline bolus
- PRBC administration

Additional controls include:

Home Button: Returns the user to the Home screen and starts a new scenario.

Recommendations/Settings Button: Allows users to:

- Review the recommendations table
- Edit patient information

- Update calculated recommendations
- Enable or disable Apgar reminders
- Enable or disable audio prompts

These settings may be modified at any point during the simulation.

8. Resuscitation Log:

Selecting Stop or Stop and View Resuscitation Log generates an editable PDF titled Resuscitation Log. The report summarizes all recorded events during the simulation and may be saved locally or shared by email or messaging applications. It has 6 sections.

- i. APGAR Scores: Displays recorded APGAR scores, which can be reviewed and edited as needed.
- ii. Interventions: Provides a chronological record of all interventions performed during the simulation. Airway details, including endotracheal tube size, insertion route, and insertion depth, can be edited.
- iii. Heart Rate: Displays heart rate documentation in either **table** or **graph** format. When viewed as a table, heart rate entries can be added, edited, or deleted.
- iv. Epinephrine Administration: Records all intratracheal and intravenous epinephrine doses administered during the simulation.
- v. FiO₂ and SpO₂: Displays FiO₂ and SpO₂ documentation in either table or graph format. Table entries can be added or edited, and the graph supports zooming for closer review.
- vi. Scribe Notes: Contains editable free-text notes recorded during the simulation.

9. Customization:

Users may customize:

- Team role assignments
- Apgar score reminders
- Audio prompts
- PPV audio guidance

- Chest compression audio guidance

10. Safety Information and Limitations

NRP Assist is intended solely as an educational support tool during NRP simulation.

Users should be aware of the following limitations:

- The application does not replace NRP certification, clinical judgment, or institutional protocols.
- It should not be used as the sole basis for clinical decision-making.
- The application is intended **only for simulation** and is not designed for use during real neonatal resuscitation.
- The generated Resuscitation Log is intended for documentation of simulation scenarios only.

Data Privacy: The app does not collect, transmit, or store any identifiable personal data. No patient identifiers are entered except for the time of birth, estimated fetal weight and gestational age, which in this case will be of the simulation scenario.

11. Device Requirements

- Compatible Apple iPhone or iPad
- iOS operating system
- Internet connection required only for installation and updates

12. Evidence Base

The NRP Assist application is based on the textbook of Neonatal Resuscitation, 9th Edition (2025) and the current American Academy of Pediatrics (AAP) and American College of Obstetricians and Gynecologists (ACOG) recommendations for neonatal resuscitation and Apgar scoring.

Appendix A — Acronyms

- NRP — Neonatal Resuscitation Program

- PPV — Positive Pressure Ventilation
- CPAP — Continuous Positive Airway Pressure
- CC – Chest Compressions
- *Alternate airway* — Endotracheal intubation or laryngeal mask placement

Appendix B — APGAR Scoring Table²

Indicator	0 point	1 point	2 points
Activity (muscle tone)	Floppy	Flexed limbs	Active
Pulse rate	Absent	<100 per minute	>100 per minute
Grimace (reflex irritability)	Absent	Minimal response to stimulation	Prompt response to stimulation
Appearance (skin color)	Blue or pale	Pink body with blue extremities	Pink
Respiration	Absent	Slow and irregular	Vigorous cry

Appendix C — References

1. Weiner, G. M., Lee, H. C., & Cooper, C. M. (Eds.). (2025). *Textbook of neonatal resuscitation* (9th ed.). American Academy of Pediatrics.
<https://doi.org/10.1542/9781610028585>
2. AMERICAN ACADEMY OF PEDIATRICS COMMITTEE ON FETUS AND NEWBORN; AMERICAN COLLEGE OF OBSTETRICIANS AND GYNECOLOGISTS COMMITTEE ON OBSTETRIC PRACTICE. The Apgar Score. *Pediatrics*. 2015 Oct;136(4):819-22. doi: 10.1542/peds.2015-2651. PMID: 26416932.